MIL-E-46215C (PA)
18 March 1975
SUPERSEDING
MIL-E-46215B (MU)
4 October 1968

MILITARY SPECIFICATION

ELEMENT, IGNITION, ELECTRIC METAL PARTS ASSEMBLY, FOR IGNITER, ELECTRIC PRIMER, M63

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts assembly for one type of ignition element designated as Element, Ignition, Electric, for use in Igniter, Electric Primer, M63.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

*MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes (ABC-STD-105)

MIL-STD-1169 - Packaging, Packing and Marking For Shipment of Inert Ammunition Components

FSC: 1390

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DRAWINGS

U.S. ARMY PICATINNY ARSENAL

8839498 - Element, Ignition, Electric, Metal Parts Assembly

*(For meaning of asterisk (*) see 6.4)

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on drawing (dwg.) 8839498, and with all requirements specified in applicable specifications.

*3.3 Workmanship Molded Parts. - All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpeage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

*3.4 Workmanship Metal Parts. - All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required) corrosion products and other foreign matter. The cleaning

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method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

3.5 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

*4.2 Classification of inspection. - The following types of inspection shall be conducted on this item.

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

*4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of twenty-five (25) complete sets of parts and sub-assemblies (i.e., twenty-five each of every component and sub-assembly, including at least five (5) items from each cavity used to produce molded parts, and be so identified as to permit tracing each item to the cavity in which it was molded), and twenty-five (25) complete assemblies. The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

*4.3.2 Inspections to be performed. - See MIL-A-48078 and para 4.4.2, testing per para. 4.4.3, molding inspections per para. 4.4.5, and Table I specified herein.

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TABLE I

FIRST ARTICLE INSPECTION

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*	Element, Ignition, Electric, Metal Parts Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inert Parts (Dwg. 8839498)				
	Examination for defects (Metal Parts)	25(a)		3.2	4.4.2.1 thru 4.4.2.3
	Examination for Defects (Molded Parts)	5 (b)		3.2	4.4.5
	Elements (Complete Assemblies) (Dwg. 8839498)				
	Examination for defects	25		3.2	4.4.2.4
	Tests				
	Electrical resistance (dwg. 8839498)	(c)	100%	3.2	4.5.1
NOTES:	(a) 25 each and every inert component (b) From each mold cavity (c) Performed on complete assemblies				

NOTES:

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*4.3.3 Rejection. - (See MIL-A-48078)

4.4 Quality conformance inspection

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

*4.4.2 Examination. - (See MIL-A-48078). - Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.1	Electrode				8839507
					NEXT HIGHER ASSEMBLY 8839498
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Concentricity of outside diameter with large end		0.40%	3.2	Gage
Minor:					
201.	Length of large end		0.65%	3.2	Gage
202.	Length of large end and outside diameter		0.65%	3.2	Gage
203.	Total length		0.65%	3.2	Gage
204.	Outside diameter		0.65%	3.2	Gage
205.	Diameter of large end		0.65%	3.2	Gage
206.	Diameter of small end		0.65%	3.2	Gage
207.	Depth of insulator groove		0.65%	3.2	Gage
208.	Outside diameter of insulator groove		0.65%	3.2	Gage
209.	Finish improper		0.65%	3.2	Visual
210.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46215C (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.2	Cup, Ignition	1 OF 1	8839508
			NEXT HIGHER ASSEMBLY: 8839498

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A&B:	None defined				
Minor:					
201.	Concentricity of outside diameter with inside diameter		0.65%	3.2	Gage
202.	Thickness through bottom		0.65%	3.2	Gage
203.	Outside diameter		0.65%	3.2	Gage
204.	Diameter of hole, min.		0.65%	3.2	Gage
205.	Inside diameter, max.		0.65%	3.2	Gage
206.	Total length		0.65%	3.2	Gage
207.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-E-46215C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.3	Disc, Support				8839513
					NEXT HIGHER ASSEMBLY 8839498
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A&B:	None defined				
Minor:					
201.	Outside diameter, min.		0.65%	3.2	Gage
202.	Inside diameter, max.		0.65%	3.2	Gage
203.	Thickness		0.65%	3.2	Gage
204.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46215C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.4	Assembly				8839498
					NEXT HIGHER ASSEMBLY 8839498
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Bridge wire inadequately swaged as determined with 4x to 6x magnification		0.40%	3.2	Visual
132.	Component missing		0.40%	3.2	Visual
133.	Parts not firmly held together		0.40%	3.2	Manual
Minor:					
201.	Total length, max.		0.65%	3.2	Gage
202.	Outside diameter of cup, max.		0.65%	3.2	Gage
203.	Bridge wire extends beyond cylindrical surface of cup		0.65%	3.2	Visual
204.	Insulator cracked		0.65%	3.2	Visual
205.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Electrical resistance. - (see dwg. 8839498). - Major A defect - This test shall be conducted one hundred percent. Any ignition element assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.1 using equipment in accordance with 4.4.4.

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2, 4.4.5 and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.2 herein.

4.4.5 Dimensional control of molded parts. - (The requirements of this paragraph apply to those dimensions which are considered "as molded", and which do not undergo subsequent operations). In place of normal sampling and inspections associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (see Note 1) a sample of five (5) parts (as molded) from each cavity shall be inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification, when more than one cavity is used in production. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.5.1 through 4.4.5.2 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the five (5) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking,

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be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

Note 1. - In establishing the curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey 07801 ATTN: SARPA-QA-A-M. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time shall be established and approved.

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CLASSIFICATION OF DEFECTS & TESTS MIL-E-46215C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.5.1	Insulator, Upper				8839506
					NEXT HIGHER ASSEMBLY 8839498
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A&B:	None defined				
Minor:					
201.	Concentricity of inside diameter with flange diameter	(a)	0.65%	3.2	Gage
202.	Small outside diameter, max.		0.65%	3.2	Gage
203.	Flange diameter		0.65%	3.2	Gage
204.	Thickness of flange		0.65%	3.2	Gage
205.	Total length		0.65%	3.2	Gage
206.	Inside diameter, max.		0.65%	3.2	Gage
207.	Evidence of poor workmanship including presence of flash		0.65%	3.3	Visual
NOTE:	(a) Sampling in accordance with para. 4.4.5				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.5.2	Insulator, Lower				8847028
					NEXT HIGHER ASSEMBLY 8839498
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A&B:	None defined				
Minor:					
201.	Inside diameter, max.	(a)	0.65%	3.2	Gage
202.	Outside diameter, min.		0.65%	3.2	Gage
203.	Thickness through bottom		0.65%	3.2	Gage
204.	Diameter of hole		0.65%	3.2	Gage
205.	Depth to bottom		0.65%	3.2	Gage
206.	Evidence of poor workmanship including presence of flash		0.65%	3.3	Visual
NOTE:	(a) Sampling in accordance with para. 4.4.5				

NOTES:

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4.5 Test methods and procedures

4.5.1 Electrical resistance. - The igniter element assembly shall be tested for electrical resistance by applying an electrical current of 20 milliamperes maximum using the ignition cup as one terminal and the electrode as the other terminal. Observation shall be made for compliance with the applicable requirement.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing

5.1.1 Level C. - The elements shall be packaged and packed in accordance with MIL-STD-1169 requirements.

5.2 Marking. - Marking shall be in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.2 Submission of inspection equipment designs for approval (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T Dover, New Jersey 07801.

6.3 Distribution of ammunition data card. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M Dover, New Jersey 07801.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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